

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z1,Z2

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple **Z1 Z2:** $Z_1Z_2g_m + Z_1$

$$H(z) = Z_1Z_2g_m + Z_1$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \frac{R_2}{C_2R_2s+1}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_2R_1R_2s + R_1R_2g_m + R_1}{C_1C_2R_1R_2s^2 + s(C_1R_1 + C_2R_2) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_2}\sqrt{R_1}\sqrt{R_2}}{C_1R_1+C_2R_2}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{C_2}\sqrt{R_1}\sqrt{R_2}}$
bandwidth: $\frac{C_1R_1+C_2R_2}{C_1C_2R_1R_2}$
K-LP: $R_1R_2g_m + R_1$
K-HP: 0
K-BP: $\frac{C_2R_1R_2}{C_1R_1+C_2R_2}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, R_2, \infty, \infty, \infty, \infty)$

$$H(s) = R_1R_2g_m + R_1$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \frac{1}{C_2s}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_2R_1s + R_1g_m}{C_2s}$$

$$\mathbf{9.3 \quad X-INVALID-ORDER-3} \quad Z(s) = \left(R_1, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 R_1 R_2 s + R_1 R_2 g_m + R_1}{C_2 R_2 s + 1}$$

$$\mathbf{9.4 \quad X-INVALID-ORDER-4} \quad Z(s) = \left(R_1, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{C_2 s}$$

$$\mathbf{9.5 \quad X-INVALID-ORDER-5} \quad Z(s) = \left(\frac{1}{C_1 s}, R_2, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + 1}{C_1 s}$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(\frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 s + g_m}{C_1 C_2 s^2}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(\frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 R_2 s + R_2 g_m + 1}{C_1 C_2 R_2 s^2 + C_1 s}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(\frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{g_m + s (C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_1 R_2 g_m + R_1}{C_1 R_1 s + 1}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{C_2 R_1 s + R_1 g_m}{C_1 C_2 R_1 s^2 + C_2 s}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_2 R_1 R_2 g_m + C_2 R_1)}{C_1 C_2 R_1 s^2 + C_2 s}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2, \infty, \infty, \infty, \infty \right)$$

$$H(s) = \frac{R_2 g_m + s (C_1 R_1 R_2 g_m + C_1 R_1) + 1}{C_1 s}$$

9.13 X-INVALID-ORDER-13 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_2 R_1 s^2 + g_m + s (C_1 R_1 g_m + C_2)}{C_1 C_2 s^2}$$

9.14 X-INVALID-ORDER-14 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{C_1 C_2 R_1 R_2 s^2 + R_2 g_m + s (C_1 R_1 R_2 g_m + C_1 R_1 + C_2 R_2) + 1}{C_1 C_2 R_2 s^2 + C_1 s}$$

9.15 X-INVALID-ORDER-15 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \infty \right)$

$$H(s) = \frac{g_m + s^2 (C_1 C_2 R_1 R_2 g_m + C_1 C_2 R_1) + s (C_1 R_1 g_m + C_2 R_2 g_m + C_2)}{C_1 C_2 s^2}$$

10 X-INVALID-WZ

11 X-PolynomialError